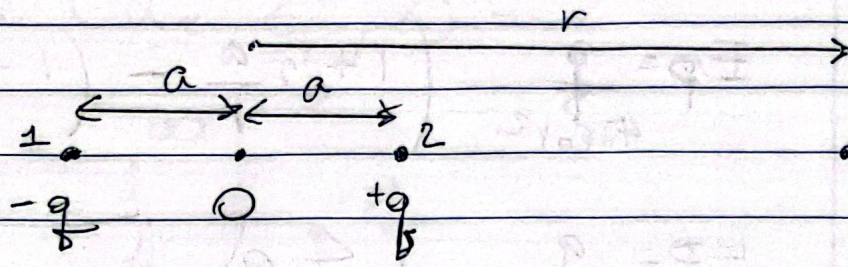


Problema #5



$$(a) \quad E_1 = \frac{1}{4\pi\epsilon_0} \frac{-q}{(r+a)^2}$$

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{+q}{(r-a)^2}$$

$$E_p = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{(r-a)^2} - \frac{q}{(r+a)^2} \right)$$

Con $r \gg a$,

$$E_p = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r^2 \left(1 - \frac{a}{r}\right)^2} - \frac{1}{r^2 \left(1 + \frac{a}{r}\right)^2} \right)$$

$$E_p = \frac{q}{4\pi\epsilon_0 r^2} \left(\left(1 - \frac{a}{r}\right)^{-2} - \left(1 + \frac{a}{r}\right)^{-2} \right)$$

Expansión
binomial:

$$\left(1 - \frac{a}{r}\right)^{-2} = 1 + 2\frac{a}{r} + 3\frac{a^2}{r^2} + 4\frac{a^3}{r^3} + \dots$$

$$\left(1 + \frac{a}{r}\right)^{-2} = 1 - 2\frac{a}{r} + 3\frac{a^2}{r^2} - 4\frac{a^3}{r^3} + \dots$$

Entonces

$$E_p = \frac{q}{4\pi\epsilon_0 r^2} \left(1 + 2\frac{a}{r} - \left(1 - 2\frac{a}{r} \right) \right)$$

$$E_p = \frac{q}{4\pi\epsilon_0 r^2} \left(\frac{4 \cdot a}{r} \right) = \frac{qa}{\pi\epsilon_0 r^3}$$

(b) $\vec{f} = q \vec{E}_p$

$$\Rightarrow \vec{f} = q \frac{qa}{\pi\epsilon_0 r^3} \hat{r}, \text{ pero } p = 2qa \Leftrightarrow \frac{p}{2} = qa$$

$$\Rightarrow \vec{f} = \frac{q}{2} \frac{p}{\pi\epsilon_0 r^3} \hat{r}, \quad p = 2E_{ion}$$

$$\vec{E}_{ion} = \frac{-Q}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\Rightarrow \vec{f} = \frac{q}{2\pi\epsilon_0 r^3} \left(-2 \frac{Q}{4\pi\epsilon_0 r^2} \right) \hat{r}$$

$$\vec{f} = \frac{-2Q^2}{8\pi^2\epsilon_0^2 r^5} \hat{r}$$

$Q^2 \rightarrow$ positivo

El menos es de atracción.

$$(c) \quad U(\alpha_e, Q, r) = ?$$

$$U = \int_r^{\infty} \vec{f} \cdot d\vec{r}$$

$$U = \int_r^{\infty} \frac{-2Q^2}{8\pi^2\epsilon_0^2 r^5} dr$$

$$U = \frac{-2Q^2}{8\pi^2\epsilon_0^2} \int_r^{\infty} r^{-5} dr = \frac{r^{-4}}{4} \Big|_r^{\infty}$$

$$U = \frac{-2Q^2}{8\pi^2\epsilon_0^2} \left(\text{---} -\frac{1}{4r^4} \right)$$

$$U = \frac{2Q^2}{32\pi^2\epsilon_0^2 r^4}$$

$$(d) \quad r_{\min} = ?$$

Por conservación del momento angular

$$(1) \quad m v_{\max} r_{\min} = m v_0 b \rightarrow v_{\max} = \frac{m v_0 b}{m r_{\min}}$$

Luego, por conservación de la energía

$$(2) \quad \frac{1}{2} m v_{\max}^2 + \frac{-2Q^2}{32\pi^2 \epsilon_0^2 r^4} = \frac{1}{2} m v_0^2$$

(1) y (2)

$$\frac{v_0^2 b^2}{r_{\min}^2} + \frac{-2Q^2}{16\pi^2 \epsilon_0^2 b^4 m} = v_0^2$$

$$\left(\frac{b^2}{r_{\min}}\right) + \frac{-2Q^2}{16\pi^2 \epsilon_0^2 b^4 m v_0^2} = 1$$

multiplicando por $\frac{r_{\min}^4}{b^4}$

$$0 = \frac{r_{\min}^4}{b^4} \cdot \frac{b^2}{r_{\min}^2} + \frac{r_{\min}^4}{b^4} + \frac{2Q^2}{16\pi^2 \epsilon_0^2 b^4 m v_0^2}$$

$$\Leftrightarrow \left(\frac{r_{\min}}{b}\right)^4 - \left(\frac{r_{\min}}{b}\right)^2 + \frac{2Q^2}{16\pi^2 \epsilon_0^2 b^4 m v_0^2} = 0$$

$$\left(\frac{r_{min}}{b}\right)^2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\left(\frac{r_{min}}{b}\right)^2 = \frac{1}{2} \left(1 \pm \sqrt{1 - \frac{2Q^2}{4\pi^2 \epsilon_0^2 b^4 m v_0^2}} \right)$$

$$r_{min}^2 = \frac{b^2}{2} \left(1 \pm \sqrt{1 - \frac{2Q^2}{4\pi^2 \epsilon_0^2 b^4 m v_0^2}} \right)$$

$$r_{min} = \pm \frac{b}{\sqrt{2}} \left(1 \pm \sqrt{1 - \frac{2Q^2}{4\pi^2 \epsilon_0^2 b^4 m v_0^2}} \right)^{1/2}$$

e)